

The numerical range is a 2-spectral set

Shanmu Jin

Department of Neurosurgery, Peking Union Medical College Hospital, No. 1 Shuaifuyuan, Dongcheng District, Beijing 100730, China
jinshanmu1129@pku.edu.cn

Abstract

For an $n \times n$ complex matrix A , let $W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}$ be its numerical range. Here x^* denotes the conjugate transpose, $\|\cdot\|_2$ the Euclidean norm, and $\|\cdot\|$ the induced operator norm. For every function f holomorphic on a neighborhood of $W(A)$, we prove that

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|$$

and consequently that $W(A)$ is a 2-spectral set for A . This proves Crouzeix's conjecture, and the constant is optimal. The central ingredient is a positive-real completion theorem relative to an auxiliary eigenbasis: an adjoint-algebra constraint on the resolvent defect of a normalized matrix-valued Carathéodory function forces the underlying matrix to have norm at most 2. The classical positive double-layer calculus supplies such completions for $f(B)$ whenever the auxiliary matrix B has simple spectrum; the eigenvalues of $f(B)$ may repeat. Sampling the associated Herglotz kernel at half the conjugate diagonal entries and at the origin cancels the completion term's contribution and leaves a comparison of two weighted Gramians; their first nonconstant terms yield the sharp estimate. Simple-spectrum approximation and convex outer approximation by domains with continuously differentiable boundary yield the general theorem.

Keywords: Crouzeix conjecture; numerical range; spectral set; matrix-valued Herglotz kernel; double-layer potential

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1. Introduction

Let $\mathbb{C}$ denote the complex field and, for $n \geq 1$, let $\mathbb{M}_n(\mathbb{C})$ denote the algebra of $n \times n$ complex matrices. For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}. \quad (1)$$

Here x^* denotes the conjugate transpose, $\sigma(A)$ denotes the spectrum, $\|x\|_2$ is the Euclidean norm, $\|X\|$ is the induced operator norm, and scalar moduli are denoted by $|\cdot|$. The set $W(A)$ is compact, the Toeplitz–Hausdorff theorem makes it convex, and $\sigma(A) \subset W(A)$. Functions holomorphic near $\sigma(A)$ are evaluated using the holomorphic functional calculus.

A compact set $K \subset \mathbb{C}$ containing $\sigma(A)$ is called a C -spectral set for A if

$$\|r(A)\| \leq C \max_{z \in K} |r(z)| \quad (2)$$

for every rational function r pole-free on K . Our main result settles Crouzeix's conjecture by identifying the sharp universal constant for $K = W(A)$.

Theorem 1.1 (Main theorem). *Let $n \geq 1$, let $A \in \mathbb{M}_n(\mathbb{C})$, and let f be holomorphic on a neighborhood of $W(A)$. Then*

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|. \quad (3)$$

Consequently, $W(A)$ is a 2-spectral set for A .

The factor 2 is optimal: for

$$N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad p(z) = z, \quad (4)$$

one has $W(N) = \{z \in \mathbb{C} : |z| \leq 1\}$ and $\|N\| = 2$.

For later use, write

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}, \quad \mathbb{T} = \partial\mathbb{D}.$$

We use I for the identity matrix of the size dictated by context, X^* for the adjoint, put $\operatorname{Re} X = (X + X^*)/2$, and let $\operatorname{alg}(T)$ denote the unital algebra generated by T . For Hermitian matrices X and Y , we write $X \preceq Y$ (equivalently, $Y \succeq X$) when $Y - X$ is positive semidefinite.

Crouzeix proved the estimate with constant 2 for 2×2 matrices and formulated the general conjecture in [1]. His subsequent dimension-free theorem gave the constant 11.08 [2]. After further structural and numerical work [3], Crouzeix and Palencia proved that $W(A)$ is a $(1 + \sqrt{2})$ -spectral set [4]. Their double-layer and Cauchy-transform argument was clarified in [5] and developed in several directions in [6,7].

The intervening literature established the conjectured constant for perturbed Jordan blocks and, more generally, scalar translates of cyclic weighted shifts, as well as for tridiagonal 3×3 matrices with elliptic numerical range centered at an eigenvalue [8–10]. Numerical and extremal investigations appear in [11,12], and a broad account of the problem and its variants is given in [13]. Subsequent work has treated Blaschke-product level sets, cyclicity reductions, compressions of shifts, further nilpotent classes, and quantitative bounds for KMS matrices [14–17]. An alternative proof for a class of weighted shifts appears in [18]. Across these cited reductions, the best dimension-free universal bound remained $1 + \sqrt{2}$.

1.1. The sharpness barrier in the symmetrized calculus

The positive boundary measure used below originates in the convex-domain integral calculus of Delyon–Delyon [19]; see also [4,20]. If B has numerical range in a convex domain Ω with continuously differentiable boundary and f is bounded by one on Ω , this calculus produces a unital positive map Φ and a holomorphic Cauchy companion g_f such that

$$2\Phi(f) = f(B) + g_f(B)^*. \quad (5)$$

Thus positivity directly controls a symmetrized functional calculus, while the target estimate concerns $f(B)$. The Crouzeix–Palencia argument couples the two terms in (5); the examples in [5] show that its abstract norm estimate is sharp at $1 + \sqrt{2}$. The corresponding general uniform-algebra theorem has the same sharp constant; its additional unital case was proposed as a possible route to 2 in [21]. Schwenninger and de Vries revisited the double-layer method [22]. Malman et al. proved that a configuration-constant refinement gives a strict improvement for every fixed numerical-range shape; they also constructed thin quadrilaterals for which the resulting bounds approach $1 + \sqrt{2}$ [23]. Separately, continuity and compactness give a constant $C_n < 1 + \sqrt{2}$ in every fixed matrix dimension n [24].

The key additional information is algebraic. For the approximants constructed in Section 5, the companion in (5) belongs to the algebra generated by B^* , where the simple-spectrum matrix B supplies

an eigenbasis that also diagonalizes $f(B)$. Treating $f(B)$ and the companion jointly preserves this coupling.

1.2. Auxiliary-basis de-symmetrization

Our argument uses the whole Cayley family associated with (5). It has four steps.

1. Applying Φ to

$$\zeta \mapsto \frac{1 + wf(\zeta)}{1 - wf(\zeta)}, \quad w \in \mathbb{D},$$

produces a matrix-valued Carathéodory function H , meaning that $H(0) = I$ and $\operatorname{Re} H(w) \succeq 0$.

2. For a simple-spectrum auxiliary matrix B , put $T = f(B)$. The double-layer identity gives the positive-real completion

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*), \quad w \in \mathbb{D}. \quad (6)$$

3. The eigenbasis of B diagonalizes $T = f(B)$ even when some values of f on the spectrum of B coincide, and it turns the unknown term in (6) into a diagonal analytic correction. We sample the Herglotz kernel of H at half the conjugate diagonal entries of T and add one vector sample at the origin. Repeated sampling points are allowed, and the origin sample cancels the diagonal correction's contribution exactly.

4. The surviving positive-matrix inequality compares two weighted Gramians. Its difference is positive, and an eigenvector argument followed by the first nonconstant Gramian term yields $\|T\| \leq 2$.

The proof uses two limits: simple-spectrum matrices tend to A , followed by outer domains converging to $W(A)$. Together they yield the general case.

2. Positive-real kernels

Let $\mathcal{K}$ be an $\mathbb{M}_n(\mathbb{C})$ -valued kernel on $\mathbb{D} \times \mathbb{D}$. We call $\mathcal{K}$ positive if, for every finite choice $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$,

$$\sum_{i,j=0}^m \xi_i^* \mathcal{K}(w_i, w_j) \xi_j \geq 0. \quad (7)$$

We shall use the following standard matrix-valued form of the Herglotz theorem. Its kernel consequence is included to fix the normalization.

Lemma 2.1 (Matrix Herglotz kernel). *Let $F: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic with $\operatorname{Re} F(w) \succeq 0$ for every $w \in \mathbb{D}$. Then*

$$\mathcal{L}_F(w, z) = \frac{F(w) + F(z)^*}{1 - w\bar{z}} \quad (8)$$

is a positive kernel.

Proof. The matrix Herglotz representation gives a positive $\mathbb{M}_n(\mathbb{C})$ -valued measure M on $\mathbb{T}$, and a Hermitian matrix J_F , such that

$$F(w) = iJ_F + \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} dM(\zeta). \quad (9)$$

For $|\zeta| = 1$, direct calculation gives

$$\frac{1}{1 - w\bar{z}} \left(\frac{\zeta + w}{\zeta - w} + \frac{\overline{\zeta + z}}{\overline{\zeta - z}} \right) = \frac{2}{(1 - w\bar{\zeta})(1 - \bar{z}\zeta)}. \quad (10)$$

The constant iJ_F cancels from (8). Substituting (9) and then summing as in (7) yields

$$2 \int_{\mathbb{T}} \left(\sum_i \frac{\xi_i}{1 - \bar{w}_i \zeta} \right)^* dM(\zeta) \left(\sum_j \frac{\xi_j}{1 - \bar{w}_j \zeta} \right) \geq 0. \quad \square$$

3. The positive-real completion principle

The following completion theorem is the core of the argument: it converts a positive-real completion constrained by the adjoint algebra of an auxiliary simple-spectrum matrix into the sharp norm bound.

Theorem 3.1 (Positive-real completion). *Let*

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}, \quad T = S \Lambda S^{-1}, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \quad (11)$$

where S is invertible, the numbers $\beta_1, \dots, \beta_n$ are distinct, and $|\lambda_i| \leq 1$ for every i . The numbers λ_i may repeat. Let $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic and satisfy

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (12)$$

and

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (13)$$

Then $\|T\| \leq 2$.

Proof. Use the simultaneous diagonalizations in (11) and put $G = S^* S \succ 0$. Since the β_i are distinct, polynomial interpolation shows that

$$\operatorname{alg}(B^*) = \left\{ (S^{-1})^* D S^* : D \text{ is diagonal} \right\}. \quad (14)$$

By (14), there is a unique diagonal analytic function Θ such that

$$H(w) - (I - wT)^{-1} = (S^{-1})^* \Theta(w) S^*.$$

Indeed,

$$\Theta(w) = S^* (H(w) - (I - wT)^{-1}) (S^*)^{-1},$$

and $\Theta(0) = 0$ because $H(0) = I$. Write $\Theta(w) = \operatorname{diag}(\theta_1(w), \dots, \theta_n(w))$. On setting $\tilde{H}(w) = S^* H(w) S$, we obtain

$$\tilde{H}(w) = G(I - w\Lambda)^{-1} + \Theta(w)G. \quad (15)$$

Since $\operatorname{Re} \tilde{H}(w) = S^* (\operatorname{Re} H(w)) S$, the real part of $\tilde{H}(w)$ is positive semidefinite. Consequently, Lemma 2.1 applies to $\tilde{H}$.

Define Hermitian matrices $P, Q, Y \in \mathbb{M}_n(\mathbb{C})$ by

$$P_{ij} = \frac{G_{ij}}{1 - \bar{\lambda}_i \lambda_j / 4}, \quad Q_{ij} = \frac{G_{ij}}{1 - \bar{\lambda}_i \lambda_j / 2}, \quad Y = Q - P. \quad (16)$$

The denominators are nonzero, and P and Q are Hermitian, because $|\lambda_i| \leq 1$. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{C}^n$, and fix $u = (u_1, \dots, u_n)^T \in \mathbb{C}^n$. In the positive-kernel inequality use the n points and vectors

$$w_i = \frac{\bar{\lambda}_i}{2}, \quad \xi_i = u_i e_i \quad (1 \leq i \leq n), \quad (17)$$

and add the point $w_0 = 0$ with vector

$$\xi_0 = v = -G^{-1}Pu. \quad (18)$$

The points w_i may coincide. Kernel positivity applies to arbitrary finite indexed lists of points and vectors, including $w_i = w_0$ when $\lambda_i = 0$. We first verify that this sample cancels the contribution of Θ to the sampled kernel quadratic form. The contribution of the second term in (15) to the kernel quadratic form is

$$2 \operatorname{Re} \sum_{i=1}^n \overline{u_i} \theta_i(w_i) \left((Gv)_i + \sum_{j=1}^n \frac{G_{ij} u_j}{1 - w_i \overline{w_j}} \right). \quad (19)$$

Because

$$1 - w_i \overline{w_j} = 1 - \overline{\lambda_i} \lambda_j / 4,$$

the parenthesis in (19) is $(Gv + Pu)_i$, which vanishes by (18).

It remains to calculate the contribution of $G(I - w\Lambda)^{-1}$. Between w_i and w_j its (i, j) entry is

$$\frac{2G_{ij}}{(1 - \overline{\lambda_i} \lambda_j / 2)(1 - \overline{\lambda_i} \lambda_j / 4)} = (4Q - 2P)_{ij}. \quad (20)$$

After compression by the coordinate vectors e_i , the two effective sample–origin blocks are $G + Q$, and the origin block is $2G$. After the cancellation of Θ 's contribution, kernel positivity gives, for the prescribed vector $v = -G^{-1}Pu$,

$$\begin{aligned} 0 &\leq \begin{pmatrix} u \\ v \end{pmatrix}^* \begin{pmatrix} 4Q - 2P & G + Q \\ G + Q & 2G \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} \\ &= u^* ((4Q - 2P) - (G + Q)G^{-1}P - PG^{-1}(G + Q) + 2PG^{-1}P)u \\ &= u^* (4Y - YG^{-1}P - PG^{-1}Y)u, \end{aligned} \quad (21)$$

Here the block quadratic form is used only on the graph $v = -G^{-1}Pu$; it is not asserted to be positive for independent u and v . The last equality uses $Q = P + Y$. Since u was arbitrary,

$$4Y - YG^{-1}P - PG^{-1}Y \succeq 0. \quad (22)$$

Balance the diagonalization by defining

$$\begin{aligned} \tilde{T} &= G^{1/2} \Lambda G^{-1/2}, & \hat{P} &= G^{-1/2} P G^{-1/2}, \\ \hat{Q} &= G^{-1/2} Q G^{-1/2}, & \hat{Y} &= \hat{Q} - \hat{P}. \end{aligned} \quad (23)$$

Expanding the scalar kernels in (16) into geometric series gives

$$\hat{P} = \sum_{k=0}^{\infty} 4^{-k} \tilde{T}^{*k} \tilde{T}^k, \quad \hat{Q} = \sum_{k=0}^{\infty} 2^{-k} \tilde{T}^{*k} \tilde{T}^k. \quad (24)$$

Indeed, $\tilde{T}^k = G^{1/2} \Lambda^k G^{-1/2}$, so the sequence $(\tilde{T}^k)$ is bounded even when some $|\lambda_j| = 1$; the geometric weights give norm convergence. It follows that

$$\hat{Y} = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k}) \tilde{T}^{*k} \tilde{T}^k \succeq 0. \quad (25)$$

Taking the congruence of (22) by $G^{-1/2}$ yields

$$4\hat{Y} - \hat{Y}\hat{P} - \hat{P}\hat{Y} \succeq 0. \quad (26)$$

Let e be an eigenvector of the Hermitian matrix $\hat{P}$, say $\hat{P}e = \alpha e$. If $\alpha > 2$, then (26) gives

$$0 \leq 2(2 - \alpha)e^* \hat{Y} e. \quad (27)$$

Since $\hat{Y} \succeq 0$ and $\alpha > 2$, (27) forces $e^* \hat{Y} e = 0$. The $k = 1$ term in (25) is $\tilde{T}^* \tilde{T}/4$, so $\tilde{T}e = 0$. The first series in (24) gives $\hat{P}e = e$, contradicting $\hat{P}e = \alpha e$ with $\alpha > 2$. The first series in (24) also begins with I , and hence

$$I \preceq \hat{P} \preceq 2I. \quad (28)$$

The $k = 1$ term of the same series now gives

$$\frac{1}{4} \tilde{T}^* \tilde{T} \preceq \hat{P} - I \preceq I.$$

Thus $\|\tilde{T}\| \leq 2$. The polar decomposition $S = UG^{1/2}$ gives

$$T = U\tilde{T}U^*,$$

where U is unitary. Therefore $\|T\| \leq 2$. $\square$

Remark 3.2. The factor $1/2$ in the sampling points (17) simultaneously produces the weights $1/2$ and $1/4$ in (24). Their positive difference bounds $\hat{P}$, and the first nonconstant term of $\hat{P}$ then gives the sharp constant. The origin sample cancels the correction's contribution by an exact algebraic identity. Distinct auxiliary eigenvalues identify $\text{alg}(B^*)$ with the diagonal algebra in the chosen basis, while the diagonal entries of T may repeat.

4. The double-layer completion

We now construct from the numerical range the function required by Theorem 3.1. The measure and companion transform used here are classical [2,4,19]. The full Cayley family yields the algebraic constraint (13) relative to the auxiliary matrix.

Lemma 4.1 (Double-layer positive map). *Let $B \in \mathbb{M}_n(\mathbb{C})$, and let $\Omega \subset \mathbb{C}$ be a bounded convex domain with continuously differentiable boundary and $W(B) \subset \Omega$. Writing $C(\partial\Omega)$ for the continuous complex-valued functions on $\partial\Omega$, there is a unital positive map*

$$\Phi: C(\partial\Omega) \longrightarrow \mathbb{M}_n(\mathbb{C})$$

such that, whenever h is holomorphic on a neighborhood of $\overline{\Omega}$,

$$2\Phi(h) = h(B) + g_h(B)^*, \quad (29)$$

where

$$g_h(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{h(\zeta)}}{\zeta - z} d\zeta, \quad z \in \Omega. \quad (30)$$

Proof. Orient $\partial\Omega$ counterclockwise. Let $\nu(\zeta)$ be the outward unit normal, let ds be arclength, and put

$$Z_\zeta = \zeta I - B, \quad \mathcal{R}_\zeta = Z_\zeta^{-1}.$$

This resolvent exists because $\sigma(B) \subset W(B) \subset \Omega$. Define the matrix-valued boundary measure

$$d\mu_B(\zeta) = \frac{1}{2\pi} \left(\nu(\zeta) \mathcal{R}_\zeta + \overline{\nu(\zeta)} \mathcal{R}_\zeta^* \right) ds. \quad (31)$$

It is positive. In fact, the supporting-line property of the convex set $\overline{\Omega}$ gives

$$\nu(\zeta)Z_\zeta^* + \overline{\nu(\zeta)}Z_\zeta = 2\operatorname{Re}(\overline{\nu(\zeta)}Z_\zeta) \succeq 0. \quad (32)$$

Indeed, for every unit vector $x \in \mathbb{C}^n$, the quadratic form on the left is

$$2\operatorname{Re}(\overline{\nu(\zeta)}(\zeta - x^*Bx)) \geq 0,$$

because $x^*Bx \in W(B) \subset \Omega$. Multiplication on the left by $\mathcal{R}_\zeta^*$ and on the right by $\mathcal{R}_\zeta$ turns the left side of (32) into the matrix in parentheses in (31).

Along the counterclockwise boundary, $d\zeta = i\nu(\zeta)ds$. The matrix Cauchy formula gives

$$\frac{1}{2\pi} \int_{\partial\Omega} \nu(\zeta)\mathcal{R}_\zeta ds = I. \quad (33)$$

Taking adjoints gives the same identity for the second layer. Hence

$$\int_{\partial\Omega} d\mu_B = 2I. \quad (34)$$

For $\varphi \in C(\partial\Omega)$, define

$$\Phi(\varphi) = \frac{1}{2} \int_{\partial\Omega} \varphi(\zeta) d\mu_B(\zeta). \quad (35)$$

By (34), Φ is unital and positive, hence $*$ -preserving; moreover, $\|\Phi(\varphi)\| \leq \|\varphi\|_\infty$, where

$$\|\varphi\|_\infty = \max_{\zeta \in \partial\Omega} |\varphi(\zeta)|.$$

For holomorphic h , the first layer of $2\Phi(h)$ is $h(B)$ by Cauchy's formula. The function g_h in (30) is holomorphic on Ω , so $g_h(B)$ is defined by the holomorphic functional calculus. To verify the relevant integral formula using only the holomorphy of g_h on Ω , choose a positively oriented contour $\Gamma \subset \Omega$ enclosing $\sigma(B)$. The compact contours Γ and $\partial\Omega$ are positively separated, so Cauchy's formula and Fubini's theorem give

$$\begin{aligned} g_h(B) &= \frac{1}{2\pi i} \int_{\Gamma} g_h(z)(zI - B)^{-1} dz \\ &= \frac{1}{2\pi i} \int_{\partial\Omega} \overline{h(\zeta)}(\zeta I - B)^{-1} d\zeta. \end{aligned} \quad (36)$$

Taking adjoints in (36) and using $d\overline{\zeta} = -i\overline{\nu(\zeta)}ds$ gives the second layer of $2\Phi(h)$. This proves (29). $\square$

Proposition 4.2 (Cayley completion for an auxiliary pair). *Let B and Ω be as in Lemma 4.1, and let f be holomorphic on a neighborhood of $\overline{\Omega}$ and satisfy*

$$\max_{\zeta \in \overline{\Omega}} |f(\zeta)| \leq 1. \quad (37)$$

Put $T = f(B)$. Then there is an analytic $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ satisfying

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (38)$$

and

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (39)$$

Moreover, $\sigma(T) \subset \overline{\mathbb{D}}$.

Proof. By (37) and spectral mapping from $\sigma(B) \subset \Omega$, we have $\sigma(T) \subset \overline{\mathbb{D}}$. For $w \in \mathbb{D}$, define on the boundary

$$c_w(\zeta) = \frac{1 + wf(\zeta)}{1 - wf(\zeta)} \quad (40)$$

and put

$$H(w) = \Phi(c_w). \quad (41)$$

The map $w \mapsto c_w$ is analytic with values in $C(\partial\Omega)$, locally uniformly on $\mathbb{D}$. Hence H is analytic, and $H(0) = I$. Furthermore,

$$\operatorname{Re} c_w(\zeta) = \frac{1 - |wf(\zeta)|^2}{|1 - wf(\zeta)|^2} \geq 0.$$

The positivity and $*$ -preservation of Φ give $\operatorname{Re} H(w) = \Phi(\operatorname{Re} c_w) \succeq 0$.

For fixed $w \in \mathbb{D}$, the same formula defines c_w holomorphically on a neighborhood of $\overline{\Omega}$: on $\overline{\Omega}$ one has $|1 - wf| \geq 1 - |w| > 0$. Hence Lemma 4.1 gives

$$2H(w) = c_w(B) + g_{c_w}(B)^*.$$

Since $T = f(B)$, the holomorphic functional calculus gives

$$c_w(B) = (I + wT)(I - wT)^{-1} = 2(I - wT)^{-1} - I.$$

Consequently,

$$H(w) - (I - wT)^{-1} = \frac{1}{2}(g_{c_w}(B)^* - I). \quad (42)$$

The function g_{c_w} is holomorphic on a neighborhood of $\sigma(B)$, so finite-dimensional holomorphic functional calculus gives $g_{c_w}(B) \in \operatorname{alg}(B)$. Thus the right-hand side of (42) belongs to $\operatorname{alg}(B^*)$, proving (39). $\square$

Corollary 4.3 (Auxiliary sharp bound). *In Proposition 4.2, if B has simple spectrum, then $\|f(B)\| \leq 2$.*

Proof. Write

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}.$$

Then

$$T = f(B) = S \operatorname{diag}(f(\beta_1), \dots, f(\beta_n)) S^{-1}.$$

Since $\sigma(B) \subset \Omega$, the bound on f gives $|f(\beta_i)| \leq 1$ for every i . Apply Theorem 3.1 to these simultaneous diagonalizations and to the function H constructed in Proposition 4.2. The values $f(\beta_i)$ may repeat. $\square$

5. Passage to arbitrary matrices and proof of the main theorem

Successive approximations pass from the auxiliary setting to arbitrary matrices.

Lemma 5.1 (Simple spectrum and numerical range). *Matrices with n distinct eigenvalues are dense in $\mathbb{M}_n(\mathbb{C})$. Moreover, for $A, B \in \mathbb{M}_n(\mathbb{C})$,*

$$W(B) \subset W(A) + \{z \in \mathbb{C} : |z| \leq \|B - A\|\}. \quad (43)$$

Proof. The discriminant of the characteristic polynomial is a nonzero polynomial in the matrix entries, as is seen at $\operatorname{diag}(1, \dots, n)$. The zero set of a nonzero complex polynomial has empty interior, so its nonvanishing set is dense in $\mathbb{M}_n(\mathbb{C})$, proving the first assertion. For a unit vector x ,

$$|x^*(B - A)x| \leq \|B - A\|,$$

which proves (43). $\square$

Lemma 5.2 (Parallel convex domains). *Let $K \subset \mathbb{C}$ be nonempty, compact, and convex. For $\delta > 0$, the set*

$$\Omega_\delta = \{z \in \mathbb{C} : \text{dist}(z, K) < \delta\}$$

is a bounded convex domain with continuously differentiable boundary, and $\overline{\Omega}_\delta = K + \delta\overline{\mathbb{D}}$.

Proof. The set Ω_δ is bounded and convex, and its closure is the Minkowski sum $K + \delta\overline{\mathbb{D}}$. Identify $\mathbb{C}$ with $\mathbb{R}^2$. The metric projection $\pi_K: \mathbb{C} \rightarrow K$ is unique and continuous. Put $d_K(z) = \text{dist}(z, K)$. For $z, h \in \mathbb{C}$, comparison first with $\pi_K(z)$ and then with $\pi_K(z + h)$ gives

$$\begin{aligned} 2 \operatorname{Re}(\overline{z + h - \pi_K(z + h)} h) - |h|^2 &\leq d_K(z + h)^2 - d_K(z)^2 \\ &\leq 2 \operatorname{Re}(\overline{z - \pi_K(z)} h) + |h|^2. \end{aligned}$$

The continuity of π_K now shows that d_K^2 is continuously differentiable, with real gradient $2(z - \pi_K(z))$. On the level set $d_K(z) = \delta$ this gradient has norm $2\delta > 0$. Since $\partial\Omega_\delta = \{d_K = \delta\}$, the implicit-function theorem shows that $\partial\Omega_\delta$ is continuously differentiable. $\square$

Proof of Theorem 1.1. Put $K = W(A)$, and let U be an open neighborhood of K on which f is holomorphic. Choose $\varepsilon_0 > 0$ such that the closed ε_0 -neighborhood of K lies in U . For $0 < \varepsilon < \varepsilon_0$, let

$$\Omega_\varepsilon = \{z \in \mathbb{C} : \text{dist}(z, K) < \varepsilon\}.$$

Lemma 5.2 makes this a bounded convex domain with continuously differentiable boundary.

By Lemma 5.1, choose simple-spectrum matrices $B_k \rightarrow A$. For fixed ε , all sufficiently large k satisfy $\|B_k - A\| < \varepsilon/2$, and then (43) gives

$$W(B_k) \subset K + (\varepsilon/2)\overline{\mathbb{D}} \subset \Omega_\varepsilon.$$

Set

$$m_\varepsilon = \max_{z \in \overline{\Omega}_\varepsilon} |f(z)|.$$

If $m_\varepsilon = 0$, then f vanishes on a neighborhood of $\sigma(A)$, so $f(A) = 0$ and there is nothing to prove. Otherwise, apply Corollary 4.3 to B_k and f/m_ε to obtain

$$\|f(B_k)\| \leq 2m_\varepsilon.$$

Continuity of the holomorphic functional calculus, obtained from a fixed contour inside U , gives $f(B_k) \rightarrow f(A)$. Hence

$$\|f(A)\| \leq 2m_\varepsilon. \tag{44}$$

Put $m_0 = \max_{z \in K} |f(z)|$. Uniform continuity of f on the closed ε_0 -neighborhood of K gives a modulus of continuity ω_f there. Since every point of $\overline{\Omega}_\varepsilon$ lies within ε of K ,

$$0 \leq m_\varepsilon - m_0 \leq \omega_f(\varepsilon).$$

Letting $\varepsilon \downarrow 0$ in (44) proves (3).

Finally, a rational function pole-free on the compact set $W(A)$ is holomorphic on a neighborhood of $W(A)$. Applying (3) to such functions proves (2) with $K = W(A)$ and $C = 2$. $\square$

6. Consequences and scope

The Hilbert-space extension follows by the standard finite-dimensional compression argument; compare [13].

Corollary 6.1 (Hilbert-space form). *Let $\mathcal{H} \neq \{0\}$ be a complex Hilbert space, with inner product linear in the first variable, and let $\mathcal{B}(\mathcal{H})$ denote its algebra of bounded operators. For $A \in \mathcal{B}(\mathcal{H})$, put*

$$W(A) = \{\langle Ax, x \rangle : x \in \mathcal{H}, \|x\| = 1\}.$$

Then, for every $p \in \mathbb{C}[z]$,

$$\|p(A)\| \leq 2 \sup_{z \in W(A)} |p(z)|. \quad (45)$$

Consequently, $\overline{W(A)}$ is a 2-spectral set for A .

Proof. The assertion is immediate if $p = 0$. Otherwise, write $d = \deg p$. For a unit vector $x \in \mathcal{H}$, let

$$\mathcal{M}_x = \text{span}\{x, Ax, \dots, A^d x\}, \quad A_x = \Pi_x A|_{\mathcal{M}_x},$$

where Π_x is the orthogonal projection onto $\mathcal{M}_x$. Induction gives

$$A_x^k x = A^k x \quad (0 \leq k \leq d), \quad (46)$$

because $A^{k+1}x \in \mathcal{M}_x$ whenever $k < d$. Moreover, $W(A_x) \subset W(A)$. Theorem 1.1, applied on the finite-dimensional space $\mathcal{M}_x$, therefore gives

$$\|p(A)x\|_{\mathcal{H}} = \|p(A_x)x\|_{\mathcal{H}} \leq \|p(A_x)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Taking the supremum over unit vectors x proves (45).

Put $K = \overline{W(A)}$. The standard inclusion $\sigma(A) \subset K$ shows that $r(A)$ is defined for every rational function r pole-free on K . Choose a closed convex neighborhood K_δ of K that contains no pole of r . Its complement is connected, so polynomial Runge approximation on K_δ , followed by the Cauchy integral formula for the functional calculus, gives polynomials p_j such that $p_j(A) \rightarrow r(A)$ and

$$\sup_{z \in W(A)} |p_j(z)| \longrightarrow \sup_{z \in W(A)} |r(z)|.$$

Passing to the limit in (45) proves the rational estimate and hence the spectral-set assertion. $\square$

The conclusion also controls polynomial and rational approximation of matrix functions. Let $A \in \mathbb{M}_n(\mathbb{C})$, let f be holomorphic on a neighborhood of $W(A)$, and let s be either a polynomial or a rational function pole-free on $W(A)$. Then $f - s$ is holomorphic on a neighborhood of $W(A)$, and Theorem 1.1 gives

$$\|f(A) - s(A)\| \leq 2 \max_{z \in W(A)} |f(z) - s(z)|. \quad (47)$$

This is the sharp universal form of the numerical-range estimate that underlies applications to GMRES and rational Krylov methods; see [7,25,26] for the role of spectral sets in those settings.

7. Discussion and further directions

Theorem 1.1 proves the scalar-valued Crouzeix conjecture. The most immediate strengthening is the uniform matrix-polynomial question: for every $m \geq 1$ and every matrix polynomial $\mathcal{F} = [f_{ij}] \in \mathbb{M}_m(\mathbb{C}[z])$, does one have

$$\|\mathcal{F}(A)\| \leq 2 \max_{z \in W(A)} \|\mathcal{F}(z)\|, \quad \mathcal{F}(A) = [f_{ij}(A)]. \quad (48)$$

Here $\|\mathcal{F}(A)\|$ is the operator norm on $(\mathbb{C}^n)^m$, whereas $\|\mathcal{F}(z)\|$ is the operator norm on $\mathbb{C}^m$. The case $m = 1$ is precisely the scalar theorem proved above; uniformity in m is an additional operator-algebraic

requirement. The foundational matrix-level disk estimate is due to Okubo and Ando [27]. Uniform matrix-level spectral-set bounds are studied in [28], and their relation to numerical radius in [29]; the constant $1 + \sqrt{2}$ at every matrix level follows from [4], while sharp estimates for ρ -contractions with $1 \leq \rho \leq 2$ are developed in [30]. The extension is known with constant 2 for weighted shift matrices [18]. For block-valued inputs, the coefficients in the completion may occupy a larger operator algebra, so the cancellation in (19) would require a genuinely operator-algebraic formulation.

The positive-real completion principle is independent of numerical ranges: it starts with a Carathéodory function whose resolvent defect lies in the adjoint algebra of an auxiliary simple-spectrum matrix sharing an eigenbasis with the matrix in the resolvent. A direct infinite-dimensional extension of Theorem 3.1, based for example on spectral measures or Riesz bases, could reveal equality cases or uniform matrix-level functional-calculus structure. The main challenge is a stable functional-calculus formulation of the auxiliary-basis correction, valid beyond finite diagonal spectral models. Related questions for unbounded numerical ranges are studied in [31], and related abstract spectral-constant problems in [32,33].

Double-layer methods beyond numerical ranges were developed in [7], while configuration-constant refinements were introduced in [23]; see also [22] for a comparison of double-layer approaches. Annular double-layer kernels are studied in [34], while [35] develops a complementary annular realization and norm theory; elliptic domains admit structured dilation models [36]. Whenever a companion transform lands in a controlled adjoint algebra, the kernel-sampling argument may replace a triangle-inequality estimate and thereby offer a route to sharp constants for other geometric spectral sets.

With the notation of the proof of Theorem 3.1, if $\|\tilde{T}\| = 2$ and $e \in \mathbb{C}^n$ satisfies $\|e\|_2 = 1$ and $\|\tilde{T}e\|_2 = 2$, then equality throughout the final Gramian inequality in the proof of Theorem 3.1 forces

$$\hat{P}e = 2e, \quad \tilde{T}^2e = 0, \quad \hat{P}\tilde{T}e = \tilde{T}e.$$

Analyzing these vector saturation conditions may classify extremal pairs (A, p) and quantify stability near the nilpotent example (4). Such a classification must allow the nonuniqueness phenomena found in [12].

Finally, O’Loughlin and Rani extend the Crouzeix–Palencia framework to scaled q -numerical ranges, $0 < |q| \leq 1$, and propose associated spectral-set problems [37]. These provide a natural testing ground for investigating a constant-2 analogue.

AI-assistance disclosure

OpenAI ChatGPT contributed the idea of sampling the matrix Herglotz kernel at half the conjugate eigenvalues and adding the origin sample that cancels the adjoint-algebra correction’s contribution; this appears in the proof of Theorem 3.1, especially (17)–(19). It also assisted with exposition, bibliography, and typesetting. The human author assumes full responsibility for the correctness of all mathematical statements and for the integrity and accuracy of all citations.

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